

Part II (4-5)

Problem 4 (10 points)

Let $X_1, \dots, X_n$ have density

$$f(x|\theta) = \begin{cases} \theta x^{\theta-1} & : 0 < x < 1, \theta > 0 \\ 0 & : \text{otherwise.} \end{cases}$$

- (a) For one observation X , write this density in exponential family form.
 (b) Let $X_1, X_2, \dots, X_n$ be an iid sample from the density in part a). Write the joint density in exponential family form and identify a sufficient statistic for θ .
 (c) Is this statistic minimal sufficient? Briefly justify.

② Exponential family form $\rightarrow f(x|\theta) = h(x) \exp(\eta(\theta) \cdot T(x) - A(\theta))$

$$\begin{aligned} f(x|\theta) &= \theta \cdot x^{\theta-1} \\ &= e^{\log[\theta x^{\theta-1}]} \\ &= e^{\log \theta + \log x^{\theta-1}} \\ &= e^{\log \theta + (\theta-1) \log x} \\ &= \exp(\log \theta + (\theta-1) \log x) \\ &= \exp\left[\underbrace{(\theta-1)}_{\eta(\theta)} \underbrace{\log x}_{T(x)} - \underbrace{(-\log \theta)}_{A(\theta)}\right] \cdot \underbrace{1}_{h(x)} \end{aligned}$$

③ $f(x|\theta) = g(T(x)|\theta) \cdot h(x)$

Steps

① Find PDF $\rightarrow f(x|\theta) = \theta \cdot x^{\theta-1} \cdot 1 \{ \theta > 0 \}$

② $f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta)$

$$= \prod_{i=1}^n \theta x_i^{\theta-1} \cdot 1 \{ \theta > 0 \}$$

$$= \left[\prod_{i=1}^n \theta \cdot x_i^{\theta-1} \right] \left[\prod_{i=1}^n 1 \{ \theta > 0 \} \right] \quad \text{③}$$

$$= \left[\theta x_1^{\theta-1} \cdot \theta x_2^{\theta-1} \cdot \dots \cdot \theta x_n^{\theta-1} \right] \left[1 \right] \quad \text{④}$$

$$\begin{aligned} &= \theta^n \left[\underbrace{(x_1 x_2 \dots x_n)^{\theta-1}}_{T(x)} \right] \left[1 \right] \\ &= \underbrace{\theta^n [T(x)]^{\theta-1}}_{g(T(x)|\theta)} \cdot \underbrace{[1]}_{h(x)} \end{aligned}$$

(*)
(A.C) Ratio approach, we assume $T(x) = T(y) \rightarrow$ Free of θ ?

$$\frac{\text{Joint PDF of } X}{\text{Joint PDF of } Y} \rightarrow \frac{f(x_1, x_2, \dots, x_n | \theta)}{f(y_1, y_2, \dots, y_n | \theta)} = \frac{\theta^n [T(x)]^{\theta-1} \cdot [1]}{\theta^n [T(y)]^{\theta-1} \cdot [1]} \quad \text{if } (*) \rightarrow = 1.$$

It's free of θ ? yes, then this is a statistic minimal sufficient.

Problem 5 (10 points) - Gamma Model

Let $X_1, \dots, X_n$ be i.i.d. with $\text{Gamma}(k, \theta)$ density

$$f(x | \theta) = \begin{cases} \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta} & : x > 0, \theta > 0 \\ 0 & : \text{otherwise.} \end{cases}$$

where $k > 0$ is known and $\theta > 0$ is unknown. (Recall: if $X \sim \text{Gamma}(k, \theta)$ with scale θ , then $\mathbb{E}[X] = k\theta$.)

- Find the method-of-moments estimator of θ .
- Derive the maximum likelihood estimator of θ .
- Let $g(\theta) = \mathbb{P}_\theta(X > c)$ for a fixed constant $c > 0$. Use the invariance property to find the MLE of $g(\theta)$.

② Steps:

① First Population Moment $\mathbb{E}[X] = k\theta$

② First Sample Moment $\frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$

③ FPM $\approx$ FSM

$$\mathbb{E}[X] \approx \bar{x}$$

$$k\theta \approx \bar{x}$$

$$k\hat{\theta} = \bar{x}$$

$$\hat{\theta} = \frac{\bar{x}}{k}$$

⑥ Steps:

① Find PDF

$$f(x|\theta) = \begin{cases} \frac{1}{\Gamma(k)\theta^k} x^{k-1} \cdot e^{-\frac{x}{\theta}} & : x > 0, \theta > 0 \\ 0 & : \text{otherwise} \end{cases}$$

② Find Joint PDF

$$f(x_1, x_2, \dots, x_n | \theta) = \prod_{i=1}^n f(x_i | \theta) = \prod_{i=1}^n \frac{1}{\Gamma(k)\theta^k} x_i^{k-1} \cdot e^{-\frac{x_i}{\theta}}$$

$$\therefore \text{lik}(\theta) = \prod_{i=1}^n \frac{1}{\Gamma(k)\theta^k} \cdot x_i^{k-1} \cdot e^{-\frac{x_i}{\theta}}$$

③ Take log likelihood

$$l(\theta) = \log(\text{lik}(\theta))$$

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$$l(\theta) = \log \prod_{i=1}^n \left[\frac{1}{\Gamma(k)\theta^k} x_i^{k-1} e^{-\frac{x_i}{\theta}} \right]$$

$$= \sum_{i=1}^n \log \left[\frac{x_i^{k-1} e^{-\frac{x_i}{\theta}}}{\Gamma(k)\theta^k} \right]$$

$$= \sum_{i=1}^n \left[\log \left[x_i^{k-1} e^{-\frac{x_i}{\theta}} \right] - \log [\Gamma(k)\theta^k] \right] = \sum_{i=1}^n \left[(k-1) \log x_i - \frac{x_i}{\theta} - \log \Gamma(k) - k \log \theta \right]$$

$$(iv) \frac{d}{d\theta} l(\theta) \rightarrow \frac{d}{d\theta} l(\theta) = 0$$

$$l'(\theta) = \sum_{i=1}^n \left[0 + \frac{x_i}{\theta^2} - 0 - k \frac{1}{\theta} \right] = \sum_{i=1}^n \left(\frac{x_i}{\theta^2} - \frac{k}{\theta} \right) = \left(\frac{x_1}{\theta^2} - \frac{k}{\theta} \right) + \left(\frac{x_2}{\theta^2} - \frac{k}{\theta} \right) + \dots + \left(\frac{x_n}{\theta^2} - \frac{k}{\theta} \right)$$

$$= \frac{1}{\theta^2} (x_1 + x_2 + \dots + x_n) - n \frac{k}{\theta} \rightarrow l'(\theta) = \frac{1}{\theta^2} \sum_{i=1}^n x_i - n \frac{k}{\theta}$$

$$\text{Now } l'(\theta) = 0$$

$$\frac{1}{\theta^2} \sum_{i=1}^n x_i = \frac{nk}{\theta} \rightarrow \frac{1}{\theta} \sum_{i=1}^n x_i = nk \rightarrow \frac{1}{n} \sum_{i=1}^n x_i = k\theta \rightarrow \bar{x} = k\theta$$

$$\hat{\theta} = \frac{\bar{x}}{k} \leftarrow \text{Same result then we got in (a)}$$

(c) First $g(\theta)$ for a single $X \sim \text{Gamma}(k, \theta)$: $g(\theta) = \int_0^{\infty} \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-\frac{x}{\theta}} dx$ $u = \frac{x}{\theta}$
 $x = \theta u, dx = \theta du$

$$g(\theta) = \int_0^{\infty} \frac{1}{\Gamma(k)} u^{k-1} e^{-u} du = \frac{\Gamma(k, \frac{\infty}{\theta})}{\Gamma(k)}, \text{ where } \Gamma(k, z) \text{ is the upper incomplete gamma function.}$$

Now apply the invariance property: the MLE of $g(\theta)$ is $g(\hat{\theta}_{MLE})$ since $\hat{\theta}_{MLE} = \frac{\bar{x}}{k}$

$$\hat{g}(\theta)_{MLE} = g\left(\frac{\bar{x}}{k}\right) = \int_0^{\infty} \frac{1}{\Gamma(k) \left(\frac{\bar{x}}{k}\right)^k} x^{k-1} \exp\left(-\frac{x}{\bar{x}/k}\right) dx = \frac{\Gamma(k, \infty / (\bar{x}/k))}{\Gamma(k)}$$

$$\hat{g}(\theta)_{MLE} = \frac{\Gamma(k, \frac{\infty k}{\bar{x}})}{\Gamma(k)}$$